Total No. of Questions : 8] | Total No. of Printed Pages : 4

Paper Code : 21304
F-404
B.C.A. (First Semester)
Examination, 2024-25
(New Course)

Paper - BCA-104-N

MATHEMATICS - I

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) Show that $\lim_{x\to 0}\left(\frac{\tan x}{x}\right)^{\frac{1}{x^3}}$ does not exist.

   (b) A function f(x) is defined as:

$$f(x)=\begin{cases}\frac{x^2}{a}-a & : x<a\\ 0 & : x=a\\ a-\frac{a^2}{x} & : x>a\end{cases}$$

Show that the function f(x) is continuous at x=a.

2. (a) If $x=\sin\left\{\frac{(\log y)}{a}\right\}$, prove that:

   $(1-x^2)y_{n+2}-(2n+1)y_{n+1}x-(n^2+a^2)y_n=0$

   (b) Evaluate $\int_0^{\pi/2}$ log sin x dx.

3. (a) Solve :

   $(D^2+D+1)y=\sin 2x$

   (b) Evaluate:

$$\frac{xdx+ydy}{xdy-ydx}=\sqrt{\frac{a^2-x^2-y^2}{x^2+y^2}}$$

4. (a) Obtain the reduction formula for $\int \sec^n x\,dx$.

   (b) Find n^{th} derivative of $\tan^{-1}\frac{x}{a}$ with respect to x.

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5. (a) Show that the function f defined as-

$$f(x) = \begin{cases} 0 & ; \quad x = 0 \\ \frac{1}{2} - x & ; \quad 0 < x < \frac{1}{2} \\ \frac{1}{2} & ; \quad x = \frac{1}{2} \\ \frac{3}{2} - x & ; \quad \frac{1}{2} < x < 1 \\ 1 & ; \quad x = 1 \end{cases}$$

has 3 points of discontinuity and also find them. https://www.mjpruonline.com

(b) State and prove Jacobian theorem.

6. (a) If $u = \tan^{-1} \frac{xy}{\sqrt{1 + x^2 + y^2}}$ show that:

$$\frac{\partial^2 w}{\partial x \partial y} = \frac{1}{(1 + x^2 + y^2)^{3/2}}$$

(b) Show that the vectors $\vec{a} = 3\hat{i} - 2\hat{j} + \hat{k}, \vec{b} = \hat{i} - 3\hat{j} + 5\hat{k}$ and $\vec{c} = 2\hat{i} + \hat{j} - 4\hat{k}$ form a right angled triangle.

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7. (a) Find the eccentricity and co-ordinate of the foci of the curve $3x^2 - y^2 = 4$

(b) Find the center, vertices, foci eccentricity and latus rectum of the hyperbola.

$9x^2 - 16y^2 - 18x - 64y - 199 = 0$

8. (a) Show that:

curl (grad r^n) = 0

(b) Prove that:

$$\nabla^2 (r^n \vec{r}) = n(n + 3) r^{n-2} \vec{r}$$

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